Answer Key - Chemistry - Graduate Level Assessment 4 (Answers are ordered exactly as in the question paper.)

Multiple Choice

1. **Answer:** A
   Options: A) 0.5; B) 2.5; C) 1.25; D) 1.5; E) 4
   Note: Correct option: A - 0.5

2. **Answer:** E
   Options: A) 0.031 g; B) 5.55 g; C) 0.021 g; D) 0.511 g; E) 0.111 g
   Note: Correct option: E - 0.111 g

3. **Answer:** A
   Options: A) 37.95 days; B) 50.6 days; C) 8 days; D) 5.6 days; E) 60.4 days
   Note: Correct option: A - 37.95 days

4. **Answer:** A
   Options: A) The 2 p peak has the lowest energy.; B) The 3 d peak has the lowest energy.; C) The 2 d peak has the lowest energy.; D) The 1 d peak has the lowest energy.; E) The 1 p peak has the lowest energy.
   Note: Correct option: A - The 2 p peak has the lowest energy.

5. **Answer:** E
   Options: A) 0.50 volts; B) 0.76 volts; C) 1.2 volts; D) 0.33 volts; E) .85 volt
   Note: Correct option: E - .85 volt

True / False

6. **Answer:** False
   Options: A) True; B) False
   Note: LLM-generated false statement. Correct answer: 'trigonal pyramidal'.

7. **Answer:** False
   Options: A) True; B) False
   Note: LLM-generated false statement. Correct answer: 'Q = 9012'.

8. **Answer:** False
   Options: A) True; B) False
   Note: LLM-generated false statement. Correct answer: '(100) / (27)'.

9. **Answer:** True
   Options: A) True; B) False
   Note: LLM-generated true statement based on MCQ.

10. **Answer:** True
    Options: A) True; B) False
    Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** 4. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
    Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** 1.8 nm. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
    Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key